

Clase # 44 - Mecánica Teórica

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1. Clase del día de hoy

Tenemos nosotros:

$$\vec{L} \equiv \vec{r} \times \vec{p}$$

Tenemos:

$$[L_i, L_j] = \epsilon_{ijk} L_k$$

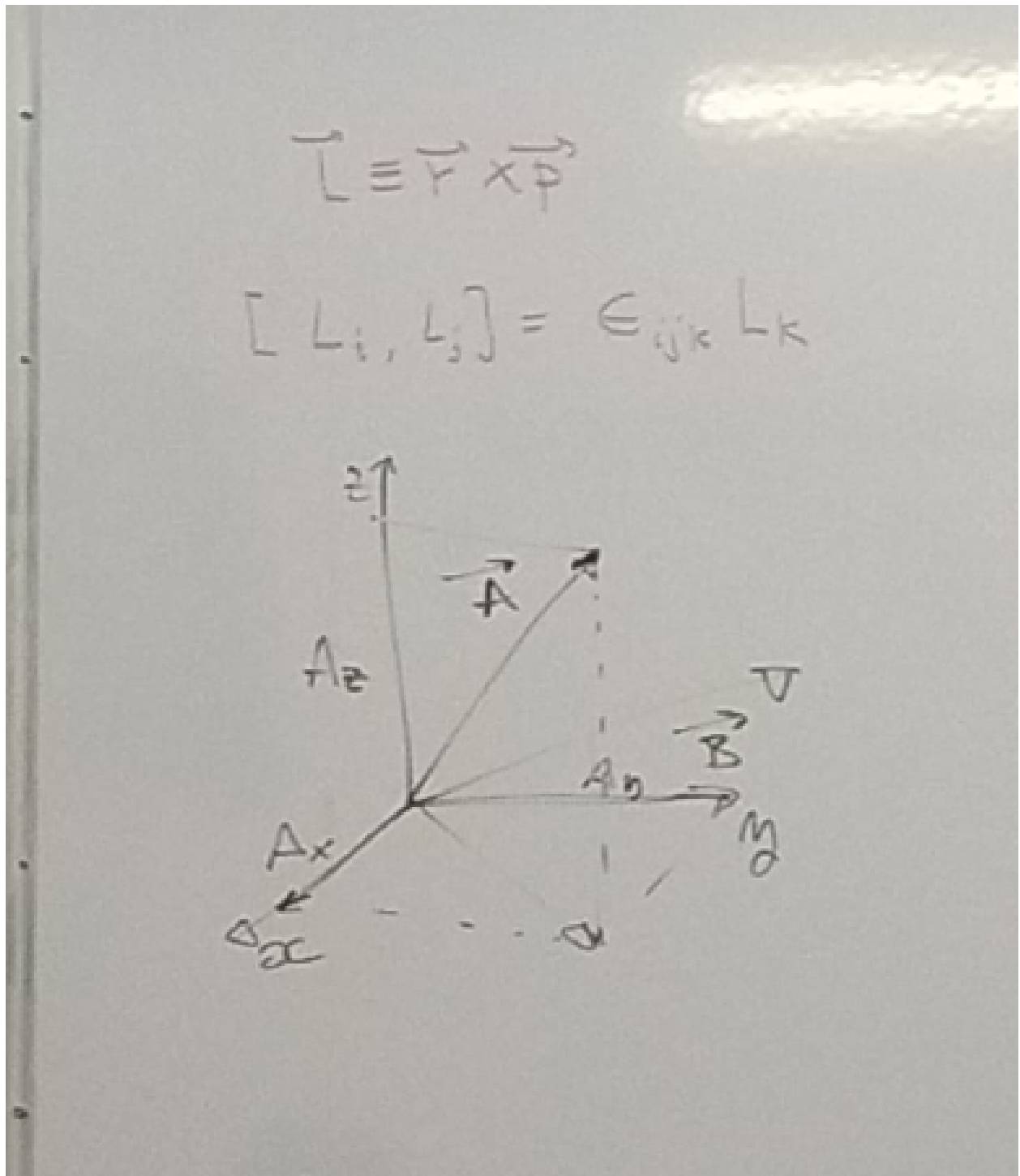


Figure 1

Tenemos:

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$$

$$\text{vec} B = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}$$

Tenemos:

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Tenemos:

$$= A_1 B_1 + A_2 B_2 + A_3 B_3$$

$$= \sum_{i=1}^3 A_i B_i = A_i B_i$$

Tenemos:

$$\vec{A} \cdot \vec{B} = A_i B_i, \quad i = 1, 2, 3$$

Tenemos:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad (1)$$

$$= \hat{x}(A_y B_z - A_z B_y) - \hat{y}(A_x B_z - A_z B_x) + \hat{z}(A_x B_y - A_y B_x)$$

Tenemos:

The image shows handwritten mathematical derivations on a piece of paper. On the left side, the components of the cross product are listed: $(\vec{A} \times \vec{B})_1 = A_2 B_3 - A_3 B_2$, $(\vec{A} \times \vec{B})_2 = A_3 B_1 - A_1 B_3$, and $(\vec{A} \times \vec{B})_3 = A_1 B_2 - A_2 B_1$. Below these, the dot product is shown as $\sum_{i=1}^3 A_i B_i = A_i B_i$. At the bottom, the components of the cross product are grouped: $(\vec{A} \times \vec{B}) \cdot \hat{x} = (\vec{A} \times \vec{B})_x = A_y B_z - A_z B_y$, $(\vec{A} \times \vec{B})_1 = A_2 B_3 - A_3 B_2$, and $(\vec{A} \times \vec{B})_y = A_z B_x - A_x B_z$. On the right side, the dot product is written as $\vec{A} \cdot \vec{B} = A_i B_i$ with $i = 1, 2, 3$. Below this, the cross product is written as a determinant: $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$. This is followed by the expanded form: $= \hat{x}(A_y B_z - A_z B_y) - \hat{y}(A_x B_z - A_z B_x) + \hat{z}(A_x B_y - A_y B_x)$.

Figure 2

Tenemos:

$$(\vec{A} \times \vec{B})_i = \epsilon_{ijk} A_j B_k$$

$$(\vec{A} \times \vec{B})_1 = \epsilon_{ijk} A_j B_k$$

Tenemos:

$$\epsilon_{123} = 1, \epsilon_{213} = -\epsilon_{123}$$

Siguiendo estas reglitas podremos desmostrar:

Cositas importantes:

$$\vec{A} \cdot \vec{B} = A_i B_i$$

$$(\vec{A} \times \vec{B})_i = \epsilon_{ijk} A_j B_k$$

$$L_i = (\vec{r} \times \vec{p})_i = \epsilon_{ijk} x_j p_k$$

Entonces

Cada vez que aparecen dos índices repetidos existe una sumatoria.

Tenemos:

$$L_i = (\vec{r} \times \vec{p})_i = \epsilon_{isl} x_s p_l$$

$$L_j = (\vec{r} \times \vec{p})_j = \epsilon_{jnm} x_n p_m$$

Tenemos:

$$[L_i, L_j] = [\epsilon_{isl} x_s p_l, \epsilon_{jnm} x_n p_m]$$

$$= \epsilon_{isl} \epsilon_{jnm} [x_s p_l, x_n p_m]$$

Tenemos:

$$= \epsilon_{isl} \epsilon_{jnm} \{x_n [x_s p_l, p_m] + [x_s p_l, x_n] p_m\}$$

$$\epsilon_{isl} \epsilon_{jnm} \left\{ x_n \underbrace{x_s [p_l, p_m]}_{=0} + x_n \underbrace{[x_s, p_m] p_l}_{=\delta_{sm}} + x_s \underbrace{[p_l, x_n] p_m}_{=-\delta_{ln}} + \underbrace{[x_s, x_n] p_l p_m}_{=0} \right\}$$

Corchetes fundamentales de Poisson

$$[L_i, L_j] = \epsilon_{isl} \epsilon_{jnm} \left\{ \underline{\delta_{sn}} x_n p_l - \delta_{ln} x_s p_m \right\} =$$

$$= \epsilon_{isl} \epsilon_{jns} x_n p_l$$

$$- \epsilon_{isl} \epsilon_{jlm} x_s p_m =$$

$$[L_i, L_j] = [\underbrace{\epsilon_{isl} x_s p_l}, \underbrace{\epsilon_{jnm} x_n p_m}]$$

$$[x_i, x_j] = 0; [p_i, p_j] = 0.$$

$$[x_i, p_j] = \delta_{ij}$$

Figure 3

Tenemos:

$$\epsilon_{isl}\epsilon_{jns} = -\epsilon_{sil}\epsilon_{sjn}$$

$$= -[\delta_{ij}\delta_{ln} - \delta_{in}\delta_{lj}]$$

Si nosotros tenemos lo siguiente (**primer término**):

$$\epsilon_{isl}\epsilon_{jns} \times_n p_\ell$$

$$= -\times_n p_\ell [\delta_{ij}\delta_{ln} - \delta_{in}\delta_{lj}]$$

$$= -\times_n p_n \delta_{ij} + \times_i p_j$$

Tenemos:

$$\epsilon_{isl}\epsilon_{jlm} \times_s p_m = -\epsilon_{lis}\epsilon_{ljm} \times_s p_m$$

$$= -\times_s p_m [\delta_{ij}\delta_{sm} - \delta_{im}\delta_{sj}]$$

$$= -\times_m p_m \delta_{ij} + \times_j p_i$$

Tenemos:

$$[L_i, L_j] = -\times_m p_m \delta_{ij} + \times_i p_j + \times_m p_m \delta_{ij} - \times_j p_i$$

Tenemos:

$$\vec{A} \times \vec{B} = A_i B_i = A_n B_n = A_m B_m$$

Llegamos a:

$$[L_i, L_j] = \epsilon_{isl} \epsilon_{jnm} \left\{ \underline{\delta_{sn}} x_n p_l - \delta_{ln} x_s p_m \right\}$$

$$= \epsilon_{isl} \epsilon_{jns} x_n p_l - \epsilon_{isl} \epsilon_{jlm} x_s p_m$$

$$[L_i, L_j] = - \overbrace{x_n p_n}^{(\vec{r} \cdot \vec{p})} \delta_{ij} + x_i p_j + x_m p_m \delta_{ij} - x_j p_i$$

$$[L_i, L_j] = x_i p_j - x_j p_i$$

Figure 4

Tenemos:

$$\begin{aligned} \epsilon_{ijk} L_k &= \epsilon_{ijk} \epsilon_{ksl} X_s P_l \\ &= \epsilon_{kij} \epsilon_{ksl} X_s P_l \\ &= (\delta_{is} \delta_{jl} - \delta_{il} \delta_{js}) X_s P_l \\ &= x_i p_j - x_j p_i \\ [L_i, L_j] &= \epsilon_{ijk} L_k \end{aligned}$$

Figure 4